

SKILL (S7 & S8)

SESSION – 7

TASK 1: Process Synchronization and Child Process Monitoring Using waitpid()

```
koumudi@Koumudi:~/OSSP/SKILL/Session_6$ cd ..
mkdir Session_7
cd Session_7
pwd
/home/koumudi/OSSP/SKILL/Session_7
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ touch waitpid_demo.c
nano waitpid_demo.c
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ gcc waitpid_demo.c -o waitpid_demo
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ ./waitpid_demo
Parent process started. PID = 2952
Parent waiting for child PID = 2953
Child process started. PID = 2953
Child process completed.
Child exited normally with status: 0
Parent process completed.
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ |
```

TASK 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
Parent process completed.
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ touch path_search.c
nano path_search.c
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ gcc path_search.c -o path_search
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ ./path_search
Enter command to locate: ls
Executable found: /usr/bin/ls
Executable permissions: YES
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ ./path_search
Enter command to locate: abcdef
Command 'abcdef' not found in PATH.
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ |
```

SESSION – 8

TASK 1: To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic

```
koumudi@Koumudi:~/OSSP/SKILL/Session_7$ cd ..
koumudi@Koumudi:~/OSSP/SKILL$ mkdir Session_8
cd Session_8
pwd
/home/koumudi/OSSP/SKILL/Session_8
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ touch variable_expand.c
nano variable_expand.c
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ gcc variable_expand.c -o variable_expand
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ ./variable_expand
Enter text: Hello $USER
Expanded result: Hello koumudi
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ ./variable_expand
Enter text: Home: $HOME
Expanded result: Home: /home/koumudi
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ ./variable_expand
Enter text: Value: $DOES_NOT_EXIST
Undefined variable: DOES_NOT_EXIST
Expanded result: Value:
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ |
```

TASK 2: To Identify Built-ins, Create Dispatch Table, Execute In- Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```
koumudi@Koumudi:~$ cd ~/OSSP/SKILL/Session_8
koumudi@Koumudi:~/OSSP/SKILL/Session_8$ ./builtin_dispatch
mini-shell> pwd
/home/koumudi/OSSP/SKILL/Session_8
mini-shell> cd ..
mini-shell> pwd
/home/koumudi/OSSP/SKILL
mini-shell> cd Session_8
mini-shell> pwd
/home/koumudi/OSSP/SKILL/Session_8
mini-shell> hello
Unknown command: hello
mini-shell> |
```